

CO2_AT1_Database Design and Connectivity

Question 1:

Pharmacy Inventory Management System

A pharmacy wants to maintain medicine inventory using MySQL. Design a Medicine table with suitable fields such as Medicine ID, Medicine Name, Manufacturer, Quantity, and Price. Write a Python program to connect to the database, create the table, insert sample records, update the stock quantity after sales, and display the available inventory.

SQL:

```
CREATE DATABASE PharmacyDB;
USE PharmacyDB;
CREATE TABLE Medicine(
    Medicine_ID INT PRIMARY KEY,
    Medicine_Name VARCHAR(100),
    Manufacturer VARCHAR(100),
    Quantity INT,
    Price DECIMAL(10,2)
);
```

Python:

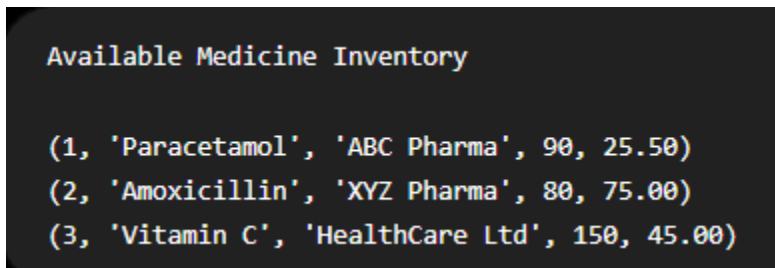
```
import mysql.connector
conn=mysql.connector.connect(host="localhost",user="root",password="irsanugP
@2005",database="PharmacyDB")
cursor=conn.cursor()
cursor.execute("""CREATE TABLE IF NOT EXISTS Medicine(
Medicine_ID INT PRIMARY KEY,
Medicine_Name VARCHAR(100),
Manufacturer VARCHAR(100),
Quantity INT,
Price DECIMAL(10,2))""")
cursor.execute("INSERT INTO Medicine VALUES (1,'Paracetamol','ABC
```

```

Pharma',100,25.50)")
cursor.execute("INSERT INTO Medicine VALUES (2,'Amoxicillin','XYZ
Pharma',80,75.00)")
cursor.execute("INSERT INTO Medicine VALUES (3,'Vitamin C','HealthCare
Ltd',150,45.00)")
conn.commit()
cursor.execute("UPDATE Medicine SET Quantity=Quantity-10 WHERE
Medicine_ID=1")
conn.commit()
cursor.execute("SELECT * FROM Medicine")
for row in cursor.fetchall():
    print(row)
cursor.close()
conn.close()

```

Output Screenshot



```

Available Medicine Inventory

(1, 'Paracetamol', 'ABC Pharma', 90, 25.50)
(2, 'Amoxicillin', 'XYZ Pharma', 80, 75.00)
(3, 'Vitamin C', 'HealthCare Ltd', 150, 45.00)

```

Question 2:

Examination Result Management System

A college maintains student details and examination results in PostgreSQL. Design Student and Result tables using appropriate primary and foreign keys. Write a Python program to create the tables, insert sample records, calculate the total marks of each student using SQL queries, and display the student names along with their total marks.

SQL:

```

CREATE TABLE Student(
    Student_ID INT PRIMARY KEY,
    Student_Name VARCHAR(100),

```

```

    Department VARCHAR(50)
);
CREATE TABLE Result(
    Result_ID INT PRIMARY KEY,
    Student_ID INT,
    Subject1 INT,
    Subject2 INT,
    Subject3 INT,
    FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID)
);

```

Python:

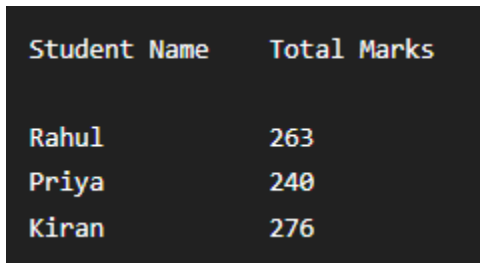
```

import psycopg2
conn=psycopg2.connect(host="localhost",database="CollegeDB",user="postgres",
password="irsanugP@2005")
cursor=conn.cursor()
cursor.execute("""CREATE TABLE IF NOT EXISTS Student(
Student_ID INT PRIMARY KEY,
Student_Name VARCHAR(100),
Department VARCHAR(50))""")
cursor.execute("""CREATE TABLE IF NOT EXISTS Result(
Result_ID INT PRIMARY KEY,
Student_ID INT,
Subject1 INT,
Subject2 INT,
Subject3 INT,
FOREIGN KEY(Student_ID) REFERENCES Student(Student_ID))""")
conn.commit()
cursor.execute("INSERT INTO Student VALUES (101,'Rahul','CSE')")
cursor.execute("INSERT INTO Student VALUES (102,'Priya','ECE')")
cursor.execute("INSERT INTO Student VALUES (103,'Kiran','IT')")
cursor.execute("INSERT INTO Result VALUES (1,101,85,90,88)")
cursor.execute("INSERT INTO Result VALUES (2,102,78,82,80)")
cursor.execute("INSERT INTO Result VALUES (3,103,92,89,95)")
conn.commit()
cursor.execute("""SELECT Student.Student_Name,
(Result.Subject1+Result.Subject2+Result.Subject3) AS Total_Marks

```

```
FROM Student JOIN Result
ON Student.Student_ID=Result.Student_ID""")
for row in cursor.fetchall():
    print(row)
cursor.close()
conn.close()
```

Output Screenshot



Student Name	Total Marks
Rahul	263
Priya	240
Kiran	276